LUPINE SCIENCE · RESEARCH BOOKLET · V1

The Savings Stack

Seven ways atomistic simulation learned to do less — the measured economics of sharing the expensive part, and the commons that makes every team faster.

July 21, 2026
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ABOUT THIS BOOKLET

Evidence before claim,
in a portable form.

This booklet is the public edition of the Savings Stack v1 release: ten citation-verified chapters on compute-savings techniques in atomistic simulation, one machine-readable measurement of cross-model anchor sharing, and the identifier audit that checked every citation. It is a research release note, not a peer-reviewed publication — every number traces to a committed record you can download, hash, and recompute.

The one-sentence version. The field independently invented seven layers of compute savings, every layer ships without a correctness certificate, and our reviewed measurement says verified gating shared across models delivers 72.4% fewer DFT evaluations.

CONTENTS

Savings Stack v1 · Released 2026-07-21 · Data record z1-union-anchor-economics.json · sha256 22c0ba51d0c798c2… (full hashes in the download pack) · Structured data ODbL-1.0 · · Not peer-reviewed; identifiers and arithmetic verified as recorded on page 14.

01 · THE ARGUMENT

Atomistic simulation spent forty years making FLOPs cheaper — and the last ten discovering that the real enemy is not the price of a FLOP but the *number of times you must call the expensive oracle at all*.

Every successful savings technology of the machine-learning era — active learning, surrogate transition-state search, Δ-learning, abstention gates — is an evaluation-count reduction with an unpriced correctness risk attached.

Our measured contribution: correctness-gated evaluation, *shared across models*, is itself a savings technology — and it scales sub-linearly. If that property generalizes, a commons in which teams contribute theorems and anchors back is a machine where we all get faster together, at no marginal cost to anyone.

That is the whole argument. The rest of this booklet is the evidence.

02 · THE COST WALL

Compute got cheap. Answers did not.

Delivered supercomputing has never been cheaper, and access is not the bottleneck for a small team — an ACCESS Explore grant (~400k credits) lands in days, with INCITE/ALCC, EuroHPC, and Director's Discretionary pools above that. The bottleneck is that serious atomistic campaigns still price out in the thousands-to-millions of oracle evaluations. A dense CI-NEB band is ~50–100 energy/force evaluations per image-set convergence accounting; a reaction network multiplies that by hundreds of paths; an honest multi-model uncertainty check multiplies again by the model count. FLOP prices falling 30× per decade does not rescue a workload whose evaluation count grows with ambition.

~30×

fall in capex per peak PFLOP in a decade — Tianhe-2 ~$7.1M/PF to El Capitan ~$0.22M/PF

$800–1,600

electricity alone per HPL-exaflop-hour on Frontier [derived] — real applications sustain 3–10× less than HPL

50–100

energy/force evaluations to converge one dense CI-NEB band, before multiplying by paths and models

10^3–10^6

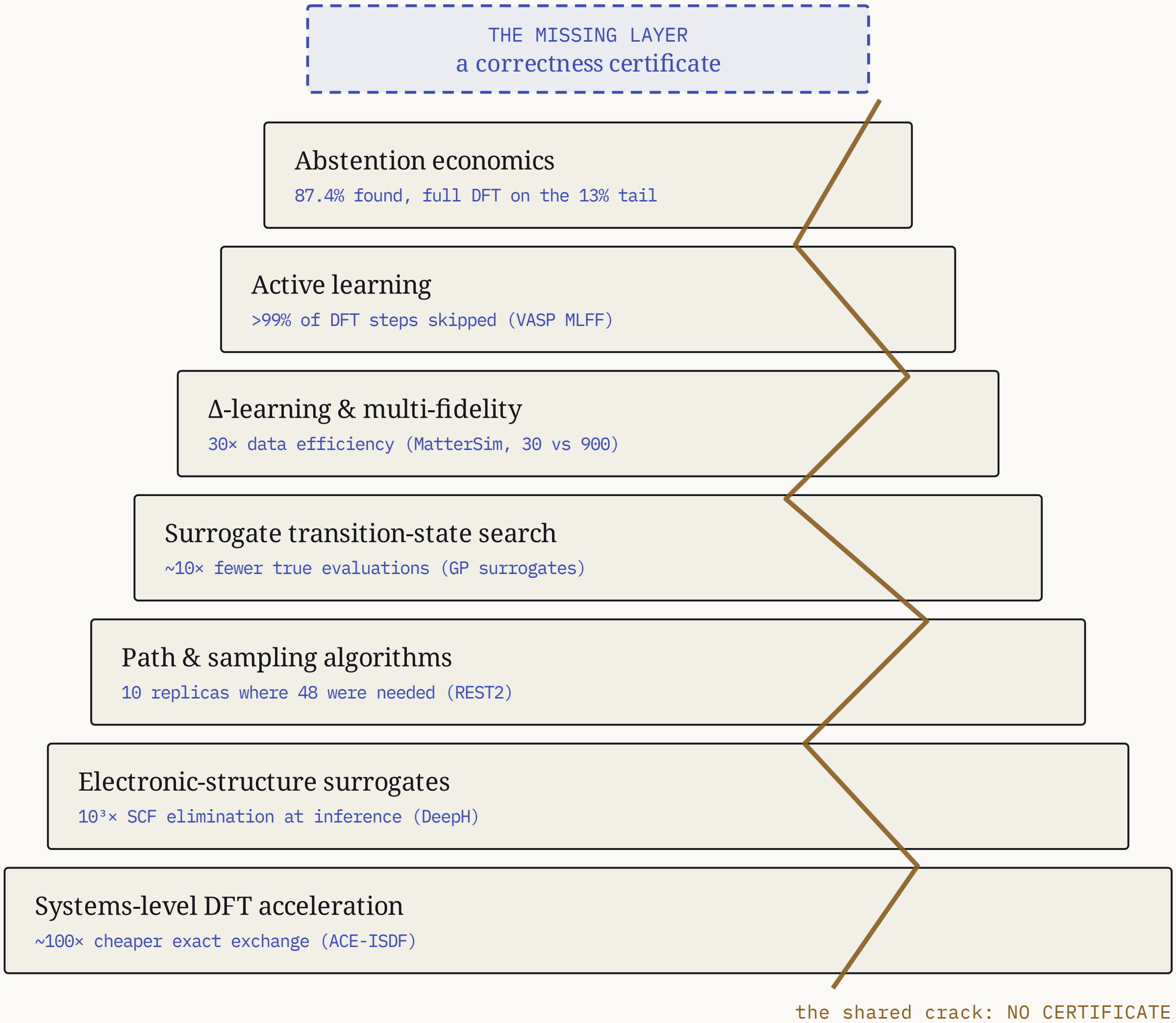
oracle evaluations for a serious campaign — the quantity every savings layer actually attacks

So the interesting question is never *“what does a FLOP cost”* but *“who lets you make fewer oracle calls — and when are they wrong.”*

Sources: backdrop chapters supercomputing-atomistic-history.md and research-compute-resourcing.md in the download pack. Machine performance figures are TOP500 HPL (Rmax). Derived figures are our arithmetic on stated inputs, labeled [derived].

03 · THE STACK, DRAWN TO SCALE

Seven layers of savings.
One crack through all of them.



Each layer is measured, replicated, and citable — the strongest verified number is drawn on its stone. And each layer’s own literature names the same hole: the trigger, the variance, the committee estimates *where the model is probably wrong*. None can say *where physics is definitely violated*.

One strongest number per layer, each traced in its digest (page 15). AdsorbML (Lan et al., npj Comput. Mater. 2023); Jinnouchi/Kresse PRB 100, 014105; arXiv:2405.04967; Garrido Torres 2019 PRL 122, 156001 (5–25×); REST2 trpcage; DeepH MoS2 supercell; ACE/ACE-ISDF 1,000-atom Si HSE.

04 · SEVEN LAYERS, ONE HOLE

What each layer saves — and what each one cannot say.

01 **Surrogate transition-state search**

5–25× fewer calls, cost decoupled from image count · Garrido Torres 2019; Koistinen/Jónsson 2017; ~10× on 500 molecular reactions (Goswami 2025)

THE HOLE
GP variance is a sampling-density signal, not an accuracy bound. Each search re-trains its surrogate, then discards it.

02 **Active learning**

>99% of first-principles steps skipped (VASP MLFF) · DP-GEN labeled 0.0044% of ~650M configurations · FLARE: ~100–250 calls

THE HOLE
Every mainstream trigger is sufficient-but-not-necessary — DP-GEN's own authors concede the committee can agree and be wrong. Rare events are the documented miss.

03 **Δ-learning, multi-fidelity, transfer**

MatterSim revPBE0-D3 water: 30 high-fidelity configurations vs 900 from scratch · DPA-2: 1–2 orders less downstream data

THE HOLE
Fine-tuning reintroduces PES softening, fails out-of-distribution, and costs a training run per model per chemistry — the wrong shape for breadth.

04 **Abstention economics**

AdsorbML: DFT-level adsorption minimum in 87.4% of ~1000 systems, full DFT only on the ~13% tail

THE HOLE
No published budget prices abstention across models — every number is per-model. The cross-model sharing economy was unmeasured territory. Until page 07.

05 **Systems-level DFT acceleration**

GPU ports 3–20× per node · reduced-rank exact exchange ~2 orders · Periodic Pulay 3× fewer SCF iterations · ML density guesses −20% total cost

THE HOLE
Cuts the cost per evaluation, not the count — and the aggressive settings carry exactly the silent-failure risk that gates exist to catch.

06 **Electronic-structure surrogates**

ML Hamiltonians eliminate 100% of SCF at inference (DeepH 10^3×; HamGNN: 4,284-atom Si in 36 s) · M-OFDFT: chemical accuracy at 27.4×

THE HOLE
Predicted Hamiltonians do not give variationally reliable energies/forces for MD or NEB — and ML functionals fail silently (DM21).

07 **Path & sampling algorithms**

Growing-string TS guess in ~20–90 gradient calls · ART nouveau: 50 vs 463 force evaluations · REST2: 10 replicas vs 48 · OPES ~10×

THE HOLE
Cheap exploration distorts or skips the TS region, and string methods converge to second-order saddles at measured rates — a wrong-

05 · THE BIG NUMBER

72.4%

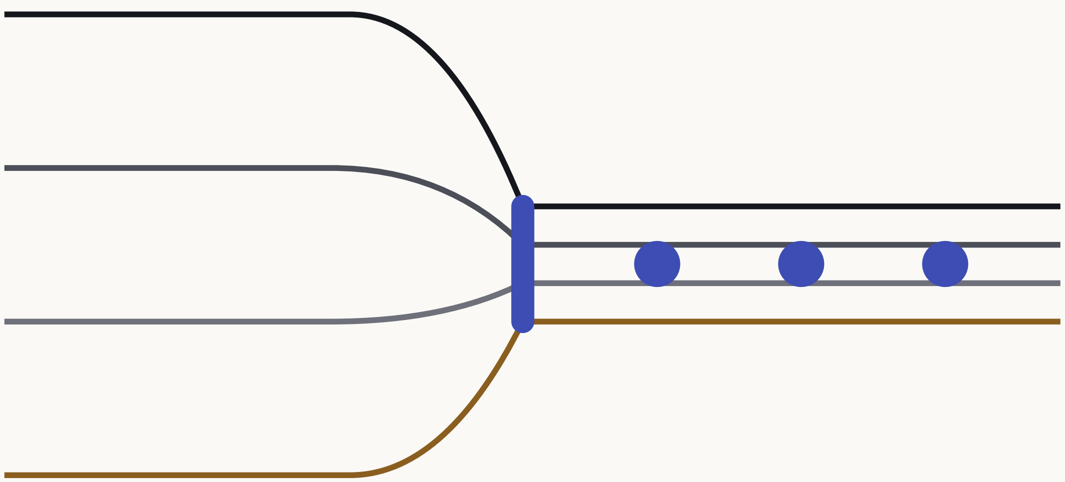
fewer DFT evaluations when four models *share their anchors* instead of each paying for its own.

The reviewed result is bounded to 29 analyzable diffusion paths on the locked Z1 panel. It was measured from recorded campaign artifacts; no new DFT was run to produce it. Detailed arithmetic remains in the hashable evidence record rather than being restated as substitute public economics.

z1-union-anchor-economics.json · sha256 22c0ba51d0c798c2… · recomputed 2026-07-21 from the Round-4 float64 campaign artifacts under the frozen anchor rule of the preregistered sparse-DFT pilot.

06 · THE CHART

Separate guidance.
Shared reference anchors.



72.4% fewer DFT evaluations on the reviewed panel.

Models largely nominate the same images, so one reference evaluation can serve multiple guidance routes. The public booklet states only the frozen reviewed result; the supporting arithmetic remains in the downloadable provenance record.

Reviewed over 29 analyzable Z1 paths. Record: z1-union-anchor-economics.json and the analysis note in the pack.

07 · THE MEASUREMENT, STATED PLAINLY

One panel, four models, and an honest denominator.

The locked Z1 panel holds 30 diffusion paths. Path 14 (`mp-756912_1_1_1_0_0`) failed CI-NEB convergence under the frozen protocol in *all four* model artifacts, so no profile exists to guide anchors — the analyzable denominator is 29 of 30, on record, with per-model failure records in the JSON. Three more paths carry partial coverage; agreement there is reported under two conventions. Nothing was imputed.

APPROVED PUBLIC RESULT

72.4% fewer DFT evaluations. Bounded to the 29 analyzable paths on the locked panel; recorded artifacts only, with no new DFT run for the review.

DO THE MODELS EVEN AGREE? PREDICTED SADDLE/EXTREMA COINCIDENCE OVER 29 PATHS

STATISTIC	AVAILABLE-MODELS	ALL-FOUR-REQUIRED
Saddle (argmax) exact	23/29	20/29
Saddle within ±1 image	26/29	23/29
Both extrema exact	10/29	9/29
Both extrema within ±1	12/29	11/29

The anchor rule is frozen: per model and path, endpoints plus the model-predicted minimum and a window around the model-predicted maximum. The naive total pays every model's set; the union total pays one evaluation per unique image per path. The protocol itself was validated before any DFT ran — sparse-protocol barrier MAE of 1.2–9.4 meV at ~7 anchors per path against ~50–100 dense evaluations — and one anchor point has since landed on real metal: 32.2 meV against a frozen 40 meV gate.

PUBLIC ECONOMICS BOUNDARY

Ratios, rounded substitutes, curve comparisons, and compounded estimates remain outside approved public copy. The downloadable record preserves the underlying measurements for review without turning them into additional public claims.

Method, reconciliation of the retracted earlier figures, and per-path tables: z1-union-anchor-economics.md in the pack. Anchor rule mirrored byte-exact from the frozen pilot implementation.

08 · THE THEOREM COMMONS

“Every team that joins makes every other team faster.”

THE THEOREM COMMONS, IN ONE SENTENCE

09 · HOW THE COMMONS COMPOUNDS

Theorems are non-rival goods with zero marginal sharing cost.

A machine-checked physical-law theorem — the curvature signature at a first-order saddle, energy–force consistency, symmetry constraints, boundary conditions — is contributed once and then gates every simulation for everyone, forever. Unlike training data, *a theorem leaks nothing about the contributor's chemistry or commercial targets*. It is the shareable residue of work a serious team does anyway: every failed simulation is a candidate theorem, and formalizing your own failure converts your most expensive knowledge into a permanent asset for a commons that pays you back in everyone else's theorems.

1. Run your campaign on the commons. Your model's known failure modes are caught by existing theorems — you save evaluations immediately.
2. Your novel failures get formalized. Each new failure mode becomes a new contributed theorem.
3. Every member's gates get sharper. False-accept and false-reject rates both fall, network-wide.
4. Savings compound. Evaluations saved grow with membership — gate quality is a monotone function of it.

The commons has two tiers, and members choose their exposure. Theorems are zero-leak by construction. Anchor libraries — a content-addressed store mapping structure-hash to DFT energy and forces — are opt-in per project: one team's anchor validates every other team's model on the same path, under the same arithmetic measured on page 07.

The reviewed union-anchor result — 72.4% fewer DFT evaluations — is the measured instance of the evaluation-side commons. Theorem layer: 271 build-locked Lean theorems in the open library, zero sorry.

10 · THE DEMAND SIDE ALREADY EXISTS

Seven digests, working independently, came back with the same hole. Read the literature's own concessions:

Learned triggers are *sufficient-not-necessary* — the committee can agree and be wrong (DP-GEN). GP variance *underestimates* error under strong extrapolation (FLARE). Ensemble σ *decorrelates* from true error. Fast exploration *skips* the transition-state region (OneOPES). String methods land on *second-order* saddles at measured rates. ML functionals fail *silently* (DM21).

Every one of those failure modes is a *physical-law violation detectable without learning anything*. A commons of formal theorems is precisely the supply for a demand the field has been circling for five years.

11 · RESTRAINT, AS A POLICY

What we do not claim.

01 Not generality beyond the measured basis.

Our measurement is one 30-path barrier panel, four universal MLIPs, one DFT engine (GPAW, frozen PBE/fd settings), one chemistry family. The union scaling curve is four points deep.

02 Not unreviewed stacking arithmetic as public economics.

Ratios, rounded substitutes, and compounded estimates are withheld from public copy unless a new source contract is reviewed.

03 Not victory over the point solutions.

For a single isolated saddle search with a cheap oracle, local GP surrogates win outright. For one stable, high-volume chemistry, fine-tuning is the right tool. Our wedge is amortized, multi-path, multi-model workloads — panels, networks, screening campaigns.

04 Not blanket validation of every literature number.

The digests retain their [UNVERIFIED] claim flags and per-source access levels. The audit verified every identifier — it does not upgrade abstract-only evidence or independently reproduce reported savings factors.

05 Not peer review.

This is a research release with committed, hashable records — an invitation to inspect and recompute, not a claim of external validation.

Restraint is the brand: a trust layer that overclaims is not one.

12 · THE CITATION AUDIT

Every identifier, checked against its registry.

All 163 unique identifiers extracted from the nine digests were verified against the official registries — arXiv IDs against the arXiv Atom API, DOI strings against Crossref Works metadata — with returned titles reviewed against the citing rows. That review caught one live but unrelated DOI hidden in a link target.

115/115	48/48	0
arXiv identifiers verified	DOI strings verified through Crossref	unresolved identifiers

Corrections applied on the record. The Chen et al. multi-fidelity citation was moved from a nonexistent DOI to Nature Computational Science 1, 46–53 (10.1038/s43588-020-00002-x). The Nandi 2021 link target — which pointed at a live but unrelated paper — was rebound to the citation text's own DOI (10.1063/5.0038301). And the VASP MLFF venue was reconciled: PRB 100, 014105 / arXiv:1904.12961 is the melting-point paper behind the >99%-bypass claim; PRL 122, 225701 is the distinct earlier hybrid-perovskite demonstration.

WHAT THE AUDIT DOES NOT DO

Identifier verification does not upgrade abstract-only evidence, does not remove each digest's [UNVERIFIED] quantitative-claim flags, and does not independently reproduce reported savings factors. It says the citations resolve to what we say they are — no more, and no less.

Full audit record: citation-verification-2026-07-21.md in the pack, with the per-identifier machine-readable artifact and the reproducible checker named there.

13 · DATA & REPRODUCTION

Download it. Hash it. Recompute it.

The Savings Stack v1 pack ships the ten chapters, the machine-readable record, the analysis script, and the audit — with a sha256 manifest over every file. Structured data is released under ODbL-1.0; prose and code keep their repository licenses; cited literature keeps its own.

savings-techniques-synthesis-2026-07-21.md	the synthesis chapter
savings-surrogate-neb.md	digest · layer 1
savings-active-learning.md	digest · layer 2
savings-delta-multifidelity.md	digest · layer 3
savings-abstention-economics.md	digest · layer 4
savings-dft-systems.md	digest · layer 5
savings-electronic-surrogates.md	digest · layer 6
savings-path-sampling-algorithms.md	digest · layer 7
supercomputing-atomistic-history.md	backdrop · the cost wall
research-compute-resourcing.md	backdrop · access & resourcing
z1-union-anchor-economics.json (+ .sha256)	the machine-readable record
z1-union-anchor-economics.md	method & reconciliation note
union_anchor_economics.py	the reproducible analysis
citation-verification-2026-07-21.md	the identifier audit
MANIFEST.sha256	hashes over every file

```
$ sha256sum -c MANIFEST.sha256          # every file: OK
$ python3 union_anchor_economics.py --local <artifacts> \
    --out z1-union-anchor-economics.json   # recompute the record
```

Pack: lupine.science/data/savings-stack-v1/ · Booklet: lupine.science/booklets/the-savings-stack.pdf · Full scientific detail: library.lupine.science, group “savings-stack”.

Evidence before claim.

The Savings Stack · v1

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Full detail — library.lupine.science · group “savings-stack”

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